

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

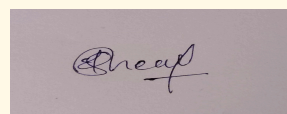
ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	Total Marks:	20 marks
CO1 Statement:	Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
Student Name:	RHEMA DAPHNE E G	Reg. No.:	192321097
Section / Batch:		Date:	20.7.2026

Code of Conduct

I ~~(Name / Reg. No)~~ **RHEMA DAPHNE E G (192321097)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- This test contains ONE question.
- Duration: 30 minutes.

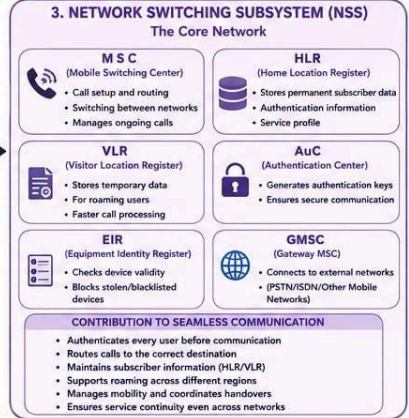
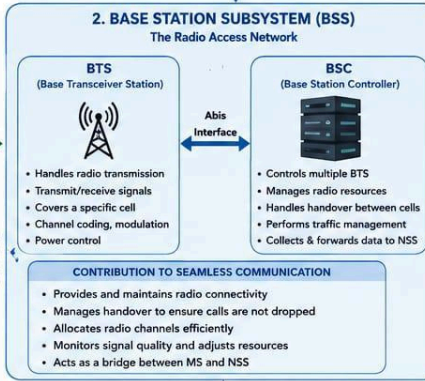
Annexure A - QUESTIONS

1. Construct a concept map that evaluates the effectiveness of the cellular network architecture. Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem. Identify the most critical components influencing communication. Justify your evaluation through meaningful concept relationships.

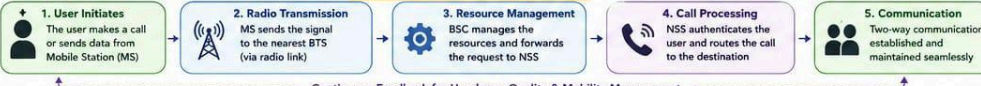
CONCEPT MAP: CONTRIBUTION OF GSM SUBSYSTEMS TOWARD SEAMLESS COMMUNICATION

SEAMLESS & CONTINUOUS COMMUNICATION

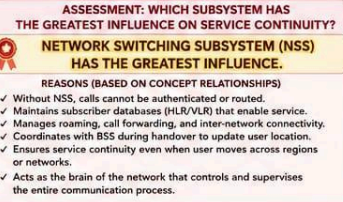
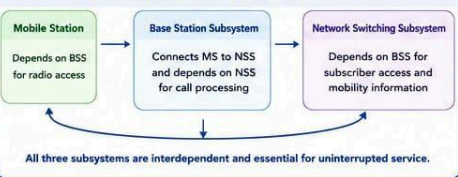
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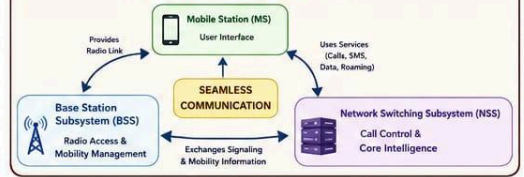
HOW THE SUBSYSTEMS WORK TOGETHER



INTERDEPENDENCIES AT A GLANCE



CONCEPT RELATIONSHIPS EXPLAINED



FINAL CONCLUSION

Mobile Station enables user communication, Base Station Subsystem provides the radio connection and manages handovers, and Network Switching Subsystem controls authentication, routing, mobility, and databases. While all are essential, the NSS has the greatest influence on service continuity because it ensures that communication is authenticated, routed, and maintained seamlessly across time, location, and networks.

KEY TAKEAWAY

Seamless communication is the result of the strong interconnection and cooperation among MS, BSS, and NSS, with NSS acting as the central coordinator that guarantees uninterrupted and reliable service.